

Supplementary Material

Evidence-Grounded RAG for G20 Climate-Policy Retrieval, Cited Synthesis and Human-Supervised Gap Screening

Detailed methods, screening rules, audit records and full matrices

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S1. G20 Membership, Entity Treatment and Corpus Cut-off

The main analysis uses 17 NDC-based analytical entities. The detailed membership and treatment table is reproduced below; the document cut-off is 1 March 2024.

Supplementary Table S1. G20 membership and NDC analytical coverage.

G20 country/entity	NDC representation used	Separate Figure 2 row?	Analytical status	Treatment and rationale
Argentina	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Australia	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Brazil	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Canada	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
China	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity

G20 country/entity	NDC representation used	Separate Figure 2 row?	Analytical status	Treatment and rationale
France	Joint EU and Member States NDC	No	Represented through EU	Within the EU NDC's geographical scope; no separate France classification produced
Germany	Joint EU and Member States NDC	No	Represented through EU	Within the EU NDC's geographical scope; no separate Germany classification produced
India	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Indonesia	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Italy	Joint EU and Member States NDC	No	Represented through EU	Within the EU NDC's geographical scope; no separate Italy classification produced
Japan	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Mexico	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Russian Federation	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Saudi Arabia	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
South Africa	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
South Korea	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
Türkiye	Updated first NDC submitted in April 2023	Yes	Assessed separately	No second NDC was available by the cut-off; evaluated using the updated first NDC
United Kingdom	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
United States	National NDC submission; latest official version available by the cut-off used for scoring	Yes	Assessed separately	Included as one national analytical entity
European Union	Joint NDC of the EU and its Member States	Yes	Assessed as one joint entity	EU-wide classification; represents the joint target, including France, Germany and Italy, but does not produce separate national results
African Union	No single AU-level NDC included in the Party-level UNFCCC corpus	No	Outside analytical scope	Permanent G20 member by the cut-off; not evaluated because the study used Party-level NDC submissions and no single AU-level NDC.

S2. Seven Policy Dimensions and Evidence Requirements

The same D1-D7 labels and evidence requirements are used across extraction, screening, reporting and supplementary files.

Supplementary Table S2. Canonical seven-dimension policy schema.

#	Policy Dimension	Evidence Extracted (Pass 1)	Scientific / policy reference (Pass 2)	Permitted Comparison and NR Rule
D1	Renewable Energy Targets	Share of renewable/non-fossil installed capacity or electricity by target year; conditionality; baseline year	Low- and zero-carbon sources supply 97–99% of global electricity by 2050 in 1.5°C pathways with no or limited overshoot and 93–97% in pathways likely to limit warming below 2°C; global electricity-sector CO ₂ reaches net zero around 2045–2055 in 1.5°C pathways.	A numerical comparison is permitted only for year- and metric-matched renewable-electricity generation shares. Capacity, non-fossil capacity, primary energy and generation share are not interchangeable. Without a documented conversion, use a Qualitative Only checklist score where evidence supports a directional judgement; otherwise classify NR.
<i>D5 interpretation: Any numerical value displayed for Loss and Damage is an ordinal checklist-derived screening score. It is not a quantified distance from an IPCC AR6 WGIII benchmark.</i>				
D2	Net Zero Pathway	Net-zero target year; gases covered (CO ₂ only or all GHGs); sectors included; conditionality	Global GHG emissions fall by about 43% from 2019 levels by 2030 in 1.5°C pathways; global net CO ₂ reaches zero around the early 2050s.	A 2030 national emissions target may be compared only with a 2030 pathway benchmark after conversion to a common baseline and emissions scope. A net-zero target year may be compared separately with the IPCC net-zero-CO ₂ timing around 2050. A 2030 percentage target must not be subtracted from a 2050 endpoint.
D3	Methane Reduction	Specific methane commitment; percentage reduction; baseline; sector scope	AR6 identifies rapid and sustained methane reductions as a critical near-term mitigation lever; a numerical comparison requires a year-, baseline- and scope-matched methane target.	A quantitative comparison requires an explicit methane target with compatible baseline, year and scope. Aggregate GHG targets are not treated as methane percentages. Where methane-specific evidence is qualitative, use the checklist; otherwise classify Absent or NR as appropriate.
D4	Adaptation Finance	Quantified finance needs or commitments; conditionality on international flows	No single country-level quantitative AR6 benchmark is used. The qualitative reference checks whether a finance need or commitment, period, mechanism and conditionality are explicit and internally distinguishable.	Finance needs, promises, allocations and actual disbursements are separate variables. They are assessed using the qualitative evidence checklist and must not be subtracted from one another as equivalent monetary commitments.

#	Policy Dimension	Evidence Extracted (Pass 1)	Scientific / policy reference (Pass 2)	Permitted Comparison and NR Rule
D5	Loss and Damage	Explicit acknowledgement of Loss and Damage impacts or needs; quantified need or contribution where stated; treatment distinct from adaptation finance; institutional, funding or operational mechanism; financing source and conditionality.	No single quantitative IPCC AR6 WGIII benchmark is used. D5 is assessed using a pre-specified qualitative climate-policy reference checklist covering explicit acknowledgement, specificity of need or contribution, distinct institutional or financial treatment, and an operational response mechanism.	All D5 cells use Qualitative Only mode. No percentage-point, monetary-shortfall or arithmetic IPCC benchmark gap is calculated. Finance values may be recorded descriptively but are not subtracted from an IPCC threshold. Where no qualifying D5 commitment is found, classify Absent. Where relevant evidence exists but is too ambiguous for a defensible checklist judgement, classify NR.
D6	Transport Sector Targets	Electrification targets or modal-shift commitments; target year; conditionality	In 1.5°C pathways, global transport-related CO ₂ emissions fall by a median 59% by 2050 relative to modelled 2020 emissions, with a 42–68% interquartile range; some illustrative pathways reach reductions of 80–90%.	A total transport-emissions target may be compared numerically only where baseline, year and scope match. EV-sales, vehicle-stock and modal-shift targets require a documented emissions model. Without one, use a Qualitative Only checklist score where defensible; otherwise classify NR. Directional electrification, EV-adoption or modal-shift language alone cannot support an Aligned classification. An Aligned D6 label requires either a compatible quantified total transport-emissions target or sufficiently complete qualitative evidence containing a defined target, target year, sectoral scope and implementation mechanism.
D7	LULUCF and Carbon Sinks	Quantified land-use commitments; offset accounting; additionality claims	AR6 supports durable enhancement of land-sector sinks and transparent accounting. Numerical comparison requires compatible land categories, baselines, boundaries and treatment of offsets.	Numerical comparison requires compatible land categories, baselines, boundaries and offset treatment. Without harmonisation, use a Qualitative Only checklist score where the evidence supports a judgement; otherwise classify NR.

S3. D5 Loss and Damage Qualitative Checklist

D5 uses Qualitative Only mode. No arithmetic IPCC WGIII benchmark gap is calculated; monetary values are descriptive evidence only.

Supplementary Table S3a. D5 evidence checklist.

Element	D5 evidence test
E1 Explicit acknowledgement	Does the NDC explicitly acknowledge Loss and Damage impacts, needs or policy treatment?
E2 Specificity	Does it state a quantified need, contribution, affected sector, geographic scope or defined action?

Element	D5 evidence test
E3 Distinct treatment	Is Loss and Damage distinguished from general adaptation or resilience language?
E4 Operational mechanism	Does it identify a fund, institution, implementation mechanism, financing arrangement or responsible process?

Supplementary Table S3b. D5 checklist-to-score mapping.

Evidence coverage	Base checklist score B
All four elements present and explicit	0.00–0.20
Three elements present	0.40
Two elements present	0.60
One element present	0.80
No qualifying D5 evidence after complete search	1.00 — Absent
Relevant but uninterpretable evidence	NR

S4. Candidate Screening Score and Class Rules

The procedure is hierarchical: commitment presence is checked first; comparability or qualitative assessability is checked second; a base score and specificity floor are then applied; conditionality is added last. For present quantitative commitments, base scores are 0.00 (meets/exceeds), 0.20 (up to 5 percentage points below), 0.40 (>5-15), 0.60 (>15-30), 0.80 (>30) and 0.90 for explicit conflict. Qualitative dimensions use the evidence-element checklist.

Specificity floors are V=0.00 for complete quantitative commitments, V=0.40 for partly specified commitments and V=0.60 for vague or aspirational language. $S0 = \max(B, V)$. Mixed conditionality adds 0.05 and fully conditional delivery adds 0.10, capped at 0.95. A score of 1.00 is reserved for Absent; NR receives no score.

Candidate classes are Aligned 0.00-0.20, Significant 0.21-0.79, Critical 0.80-0.95, Absent 1.00 and Not Directly Comparable (NR).

Supplementary Table S4. Deterministic screening rubric.

Decision stage	Condition	Numerical treatment
Presence	No relevant commitment after complete search	1.00, Absent
Comparability	Commitment exists but neither harmonisation nor a defensible qualitative judgement is possible	NR, Not Directly Comparable
Benchmark alignment	Meets or exceeds benchmark	B=0.00
Benchmark alignment	≤5 percentage-point shortfall	B=0.20
Benchmark alignment	>5–15 percentage-point shortfall or substantial partial coverage	B=0.40
Benchmark alignment	>15–30 percentage-point shortfall or material incompleteness	B=0.60
Benchmark alignment	>30 percentage-point shortfall or major omission	B=0.80
Conflict	Explicitly contradicts benchmark direction	B=0.90
Specificity	Fully quantified and scoped	V=0.00
Specificity	Partly specified	Minimum V=0.40
Specificity	Vague or aspirational	Minimum V=0.60

Decision stage	Condition	Numerical treatment
Conditionality	Unconditional	No adjustment
Conditionality	Mixed unconditional and conditional target	+0.05; minimum final score 0.25
Conditionality	Fully conditional	+0.10; minimum final score 0.40
Final cap	Present commitment	Maximum 0.95
Candidate screening outcomes	Aligned / Significant / Critical / Absent / Not Directly Comparable	0.00–0.20 / 0.21–0.79 / 0.80–0.95 / 1.00 / NR

S5. Metric and Temporal Harmonisation Rules

A numerical gap is permitted only when year, metric, denominator, unit and scope align or can be harmonised with documented inputs. Otherwise the cell is assessed qualitatively or assigned NR. Global IPCC pathway values are not equity-adjusted national targets.

Supplementary Table S5. Metric and temporal harmonisation rules.

Mismatch	Valid harmonisation	Classification where required inputs are unavailable
National emissions reduction from a non-2019 baseline vs. 2030 IPCC reduction from 2019	Convert through absolute emissions using the same gas and sectoral scope	No arithmetic gap; Qualitative Only where supported, otherwise NR
2030 national target vs. 2050 IPCC endpoint	Use an IPCC value for the same target year from an explicit pathway time series	No arithmetic gap; Qualitative Only where supported, otherwise NR
Installed capacity in GW vs. electricity-generation share	Convert using complete capacity mix and technology-specific capacity factors	No arithmetic gap; Qualitative Only where supported, otherwise NR
Installed non-fossil capacity share vs. renewable-electricity share	Convert generation by technology; exclude non-renewable non-fossil technologies where appropriate	No arithmetic gap; Qualitative Only where supported, otherwise NR
Primary-energy share vs. electricity-generation share	Use a documented energy-balance or energy-system model	No arithmetic gap; Qualitative Only where supported, otherwise NR
Aggregate GHG reduction vs. methane-specific reduction	Decompose the target by gas using compatible inventory data	No methane percentage gap; use qualitative evidence, Absent or NR as applicable
EV or modal-shift target vs. total transport-emissions reduction	Use a documented fleet and transport-emissions model	No arithmetic gap; Qualitative Only where supported, otherwise NR
Finance need vs. pledge or disbursement	No direct conversion; assess as separate qualitative evidence fields	Qualitative Only

S6. Model Backends, No-RAG Settings and Prompt Templates

Three RAG backends used common retrieval, temperature and output limits. The internally matched GPT-4o RAG/no-RAG comparison used the same recorded API configuration; the exact dated model identifier was not retained.

Supplementary Table S6. Matched GPT-4o settings.

Setting	GPT-4o without RAG	GPT-4o with RAG
Recorded model metadata	GPT-4o API configuration used in the original paired run; exact dated model identifier not recorded	Same recorded GPT-4o API configuration; exact dated model identifier not recorded
Questions	Same 80-question evaluation set	Same
Temperature	0.0	0.0
Maximum output	800 tokens	800 tokens
Retrieval	None	Hybrid top-five
External context	None	Retrieved NDC/IPCC passages
Knowledge instruction	Parametric knowledge permitted	Retrieved evidence only
Citation instruction	Do not invent citations	Cite document ID and page
Uncertainty instruction	State uncertainty	State insufficient context
Evaluation evidence	Author-prepared evidence package; same package used for both conditions	Author-prepared evidence package; same package used for both conditions
Claim judge	GPT-4o-mini automated judge; no expert-human claim adjudication	GPT-4o-mini automated judge; no expert-human claim adjudication

RAG prompt

Role: You are a climate policy analyst with access only to the supplied source evidence.

Answer using only the information contained in the retrieved context. Do not introduce unsupported information. If the context does not contain sufficient evidence, state this explicitly.

Cite factual claims using the supplied document identifier and page number.

Retrieved context:
{{TOP_5_RETRIEVED_CHUNKS}}

Question:
{{QUESTION}}

No-RAG prompt

Role: You are a climate policy analyst.

Answer the following question as accurately and specifically as possible using your internal model knowledge. No external documents, retrieved passages or document metadata are available.

Do not invent quotations, document identifiers, page numbers, source passages or citations. Where you cannot verify a specific value, target year, baseline year, country commitment or IPCC reference, state that uncertainty explicitly.

Question:
{{QUESTION}}

S7. Evaluation-Set Composition and Single-Reviewer Audit

Supplementary Table S7. Question composition by category and difficulty.

Category	Easy	Medium	Hard	Total
Factual	8	7	15	30
Comparative	2	4	19	25
Gap Detection	2	3	20	25
Total	12	14	54	80

Supplementary Table S8. Single-reviewer outcomes.

Question category	N	Full	Partial	None	Full or Partial
Factual	30	18 (60.0%)	9 (30.0%)	3 (10.0%)	27 (90.0%)
Comparative	25	2 (8.0%)	23 (92.0%)	0 (0.0%)	25 (100.0%)
Gap Detection	25	1 (4.0%)	18 (72.0%)	6 (24.0%)	19 (76.0%)
Overall	80	21 (26.3%)	50 (62.5%)	9 (11.3%)	71 (88.8%)

The resource remains author-prepared and non-adjudicated: 21 items were Full, 50 Partial and 9 None. Cohen's Kappa is not available because the author and reviewer did not independently assign the same categorical labels.

S8. Claim-Rate Definitions, Bootstrap Design and Archived Aggregate Results

GPT-4o-mini labelled atomic claims as Supported, Partial, Unsupported or Fabricated relative to author-prepared evidence packages. The pooled Unsupported/Fabricated rate is the total number of Unsupported plus Fabricated claims divided by total evaluable factual claims. Paired inference resampled question identifiers so all claims from an answer remained clustered.

Supplementary Table S8a. Reported aggregate claim-label rates.

Comparison	Condition	Judge-estimated Supported rate	Judge-estimated Partial rate	Judge-estimated Unsupported/Fabricated rate	Raw claim rows included?	Status
GPT-4o no-RAG	no_rag	0.3523	0.228	0.4197	No	Reported aggregate only; raw claim-level file unavailable
GPT-4o + RAG	rag	0.514	0.3326	0.1533	No	Reported aggregate only; raw claim-level file unavailable
LLaMA-3.1-8B + RAG	llama_rag	0.4085		0.221	No	Reported aggregate only; raw claim-level file unavailable

Supplementary Table S8b. Reported paired no-RAG/RAG results.

Metric	No RAG	RAG	RAG minus No RAG	95% CI	p-value
Judge-estimated Supported-claim rate	0.3523	0.514	0.1617	[0.085, 0.241]	<0.001
Judge-estimated Unsupported/Fabricated claim rate	0.4197	0.1533	-0.2664	[-0.344, -0.187]	<0.001
Judge-estimated Partial-claim rate	0.228	0.3326	0.1046	[0.050, 0.157]	<0.001
ROUGE-1 F1	0.3129	0.2821	-0.0308	[-0.053, -0.010]	0.005
ROUGE-2 F1	0.128	0.109	-0.019	[-0.034, -0.005]	0.011
ROUGE-L F1	0.2201	0.1875	-0.0326	[-0.051, -0.015]	<0.001
Answer coverage	0.6386	0.5585	-0.0801	[-0.115, -0.045]	<0.001

S9. Seven-Mode Diagnostic Taxonomy

Supplementary Table S9. Operational definitions for the seven diagnostic modes.

Mode	Operational definition	Root-cause layer
FM1 Retrieval Miss	The evidence required to answer the question exists in the frozen corpus, but no sufficiently relevant document or chunk appears in the top-k retrieved context.	Retrieval
FM2 Context Noise	Relevant evidence is present in the retrieved context, but excessive irrelevant, duplicated or conflicting material distracts the generator or causes an off-target answer.	Retrieved-context quality
FM3 Context Incompleteness	The correct document or some relevant evidence is retrieved, but a necessary target, baseline, year, scope, conditionality clause or supporting passage is missing, truncated or omitted from the top-k context.	Retrieved-context quality
FM4 LLM Hallucination — automated-judge flag	Adequate relevant context is available, but GPT-4o-mini labelled one or more material claims Unsupported or Fabricated relative to the author-prepared evidence package. This is an automated judge classification rather than an expert-human determination that the claim was factually false.	Generation
FM5 LLM Vagueness	Adequate evidence is available, but the answer remains overly general, omits requested numerical or policy details, or fails to provide the required comparison.	Generation
FM6 Metric Mismatch	The answer is substantively correct and source-supported, but receives a low automatic score because of paraphrasing, citation formatting, wording, unit presentation or mismatch with the author-prepared reference answer.	Evaluation artefact
FM7 Corpus Gap	The evidence required to answer the question is genuinely absent from the frozen corpus; the system cannot answer or appropriately declares insufficient context.	Corpus coverage

S10. Heatmap Review Sample and Formal Eight-Cell Metrics

The extended review covered 35 cells, but 27 cited post-cut-off evidence and were excluded from formal metrics. The formal subset consists of eight cells. Detailed row-level records are included in Archive S1.

Supplementary Table S10. Formal eight-cell performance metrics.

Metric	Value	Interpretation
Reviewed heatmap cells	8/119, 6.7%	Cut-off-compliant subset across five entities and five dimensions
Exact classification agreement / accuracy	50.0% (4/8)	System and human reviewer assigned the same final class
Classification disagreement	50.0% (4/8)	System and human reviewer assigned different classes
Macro-averaged F1	0.3333	Equal weighting across the five pre-specified outcomes
Unweighted Cohen's Kappa	0.2889	Chance-corrected system–human agreement across all five outcomes
Linear weighted Cohen's Kappa	0.2632	Ordinal agreement on seven cells, excluding the human NR case
Most frequent error	Significant → Aligned	Two of the four errors used this direction

S11. Formal Confusion Matrix and Diagnostic Review Status

Supplementary Table S11a. Five-class confusion matrix for the formal eight-cell pilot.

Human \ System	Aligned	Significant	Critical	Absent	Not Comparable	Directly Comparable
Aligned (n=0)	0	0	0	0	0	
Significant (n=6)	2	3	0	1	0	
Critical (n=1)	0	0	1	0	0	

Human \ System	Aligned	Significant	Critical	Absent	Not Comparable	Directly
Absent (n=0)	0	0	0	0	0	
Not Directly Comparable (n=1)	0	0	0	1	0	

Supplementary Table S11b. Extended 35-cell diagnostic metrics.

Metric	Value	Notes
Reviewed cells	35	Diagnostic subset; 27 rows flagged as potentially post-cut-off
Exact matches	19	
Accuracy	0.5428571428571428	Provisional diagnostic result
Macro-F1 (5 classes)	0.2655038759689922	Provisional diagnostic result
Unweighted Cohen's Kappa	0.21237693389592138	Provisional diagnostic result
Linear weighted Kappa	0.30612244897959184	Provisional; ordinal classes only

Supplementary Table S11c. Extended 35-cell diagnostic confusion matrix.

Human \ System	Aligned	Significant	Critical	Absent	Not Comparable
Aligned	0	0	0	0	0
Significant	3	16	1	2	0
Critical	0	3	1	0	0
Absent	0	2	2	2	0
Not Comparable	1	0	0	2	0

The 35-cell results are diagnostic only because 27 rows use post-cut-off evidence. They are not used in the main manuscript's formal classifier metrics.

S12. Category-Level Answer-Quality Results

Supplementary Table S12. Per-category GPT-4o+RAG evaluation results.

Category	ROUGE-1 F1	BERTScore F1	Answer Rel.
Factual (n=30)	0.3302	0.7836	0.7008
Comparative (n=25)	0.2498	0.7805	0.3559
Gap Detection (n=25)	0.2497	0.7726	0.0729

S13. Full 119-Cell Numerical Screening Matrix

D1-D5 and D7 retain the original candidate system values. D6 shows the post-hoc source-audited reporting values. The original automated D6 column is preserved in Archive S1.

Supplementary Table S13. Full numerical matrix used for reporting Figure 2.

	D1 Renewable	D2 Net Zero	D3 Methane	D4 Adapt. Finance	D5 Loss & Damage	D6 Transport	D7 LULUCF
Entity							
Argentina	0.70 Significant	0.70 Significant	0.85 Critical	0.85 Critical	0.70 Significant	0.60 Significant	0.70 Significant
Australia	0.70 Significant	0.70 Significant	0.85 Critical	0.65 Significant	1.00 Absent	1.00 Absent	1.00 Absent
Brazil	0.70 Significant	0.70 Significant	0.85 Critical	0.85 Critical	1.00 Absent	0.60 Significant	0.70 Significant
Canada	0.70 Significant	0.70 Significant	0.65 Significant	0.85 Critical	0.70 Significant	0.40 Significant	0.70 Significant
China	0.60 Significant	0.60 Significant	0.65 Significant	0.65 Significant	0.70 Significant	0.60 Significant	0.70 Significant
EU	0.60 Significant	0.75 Significant	0.65 Significant	0.65 Significant	1.00 Absent	0.40 Significant	0.70 Significant
India	0.70 Significant	0.70 Significant	0.85 Critical	0.65 Significant	1.00 Absent	1.00 Absent	0.70 Significant
Indonesia	0.70 Significant	0.70 Significant	0.70 Significant	0.65 Significant	0.00 Aligned	0.40 Significant	0.70 Significant
Japan	0.60 Significant	0.60 Significant	0.85 Critical	0.65 Significant	1.00 Absent	0.40 Significant	0.60 Significant
Mexico	0.60 Significant	0.85 Critical	0.80 Critical	0.85 Critical	1.00 Absent	0.60 Significant	1.00 Absent
Russia	1.00 Absent	0.85 Critical	0.85 Critical	0.65 Significant	1.00 Absent	1.00 Absent	1.00 Absent
Saudi Arabia	0.60 Significant	0.80 Critical	0.65 Significant	0.85 Critical	1.00 Absent	0.60 Significant	1.00 Absent
South Africa	0.60 Significant	0.70 Significant	0.85 Critical	0.65 Significant	0.00 Aligned	0.60 Significant	1.00 Absent
South Korea	0.70 Significant	0.70 Significant	0.85 Critical	0.65 Significant	0.60 Significant	0.40 Significant	0.60 Significant
Türkiye	0.60 Significant	0.85 Critical	0.70 Significant	0.65 Significant	1.00 Absent	0.40 Significant	0.60 Significant
United Kingdom	0.60 Significant	0.70 Significant	0.60 Significant	0.65 Significant	1.00 Absent	0.40 Significant	0.60 Significant
United States	0.60 Significant	0.85 Critical	0.85 Critical	0.85 Critical	1.00 Absent	0.60 Significant	0.70 Significant

S14. D6 Transport: Original Outputs and 17-Cell Source Audit

The pilot review produced 0/3 agreement on D6 and triggered a post-hoc source audit of all 17 cells. The audit removed all 11 original Aligned labels and produced 14 Significant and 3 Absent reporting outcomes. This audit is excluded from formal classifier metrics.

Supplementary Table S14. Original and source-audited D6 outputs with evidence summaries.

Entity	Original score/class	Audited score/class	Cut-off-compliant evidence summary	Source-audit rationale
Argentina	0.7 Significant	0.6 Significant	The NDC describes a 2030 Avoid-Shift-Improve approach, intermodality, active mobility, public transport, biofuels, electrification and hydrogen, but gives no quantified total transport-emissions or fleet-share target.	Directional transport mitigation evidence is sufficient for a Significant screening label, but not for Aligned or a numerical comparison with the IPCC transport pathway.
Australia	0 Aligned	1 Absent	The assessed NDC updates the economy-wide 2030 and net-zero targets but does not state an explicit transport-sector, vehicle, modal-shift or transport-emissions commitment.	Absent refers only to the assessed NDC text, not to the absence of Australian transport policy outside the NDC.
Brazil	0 Aligned	0.6 Significant	The NDC records a substantial renewable share in the transport energy mix and identifies RenovaBio as a decarbonisation instrument, but it does not provide a forward quantified total transport-emissions, EV-share or modal target.	Existing biofuel policy supports a Significant screening label, but it does not establish transport-pathway alignment.
Canada	0 Aligned	0.4 Significant	Canada had defined light-duty ZEV sales milestones of 20% by 2026, 60% by 2030 and 100% by 2035, supported by regulation and charging/transit programmes. The metric is new-vehicle sales, not total transport emissions.	The commitment is specific and operational, but partial in sector coverage and metric.
China	1 Absent	0.6 Significant	The NDC promotes green and low-carbon transport, new-energy vehicles, public transport, railway and waterborne freight, and cleaner shipping and aviation, but does not state a quantified total transport-emissions target.	The original Absent label is not supported because relevant transport commitments are present.
European Union	0 Aligned	0.4 Significant	The EU framework includes dated car and van CO2 standards and zero-emission vehicle milestones within the Fit for 55 package. These are strong road-vehicle measures but do not constitute one directly comparable total transport-emissions target.	The evidence supports Significant, not Aligned, because the benchmark and policy metrics differ and the target is not transport-wide.
India	1 Absent	1 Absent	The updated NDC does not state an explicit transport-sector emissions target, EV target, modal-shift target or transport implementation commitment.	Absent is limited to the NDC document and does not deny the existence of domestic transport programmes outside it.
Indonesia	0 Aligned	0.4 Significant	The NDC and its implementation context include dated BEV deployment targets for four-wheel and two-wheel vehicles and policies for biofuels and vehicle electrification. These are quantified deployment targets rather than a total transport-emissions target.	Specificity is high, but the measure covers vehicle deployment rather than complete transport-sector emissions.
Japan	0 Aligned	0.4 Significant	Japan's implementation plan provides a 2030 transport-sector emissions level and reduction from 2013, alongside vehicle-electrification measures. The target year and baseline differ from the IPCC 2050 reference.	This is among the strongest D6 commitments, but the paper's own comparability rule prevents a direct Aligned judgement.
Mexico	0.6 Significant	0.6 Significant	The NDC includes electric mobility, vehicle-efficiency, public-transport and modal measures, but no quantified total transport-emissions or national EV-sales target.	The original class is retained, but the score should be interpreted as checklist-based rather than an arithmetic gap.
Russian Federation	0.7 Significant	1 Absent	The NDC states an economy-wide 2030 target and broad mitigation policies but does not identify a transport-sector emissions, vehicle or modal-shift commitment.	The existing Significant label lacks a transport-specific evidence basis in the submitted NDC.
Saudi Arabia	1 Absent	0.6 Significant	The NDC identifies vehicle-efficiency standards, retirement of inefficient vehicles, mass-transit, rail and bus-network measures, but provides no quantified total transport-sector target.	The original Absent label is not supported because explicit transport mitigation measures are present.
South Africa	0 Aligned	0.6 Significant	The NDC references the Green Transport Strategy, bus rapid transit, road-to-rail shifts and preparation for vehicle decarbonisation, but does not contain a quantified 2030 transport-sector target.	The evidence does not support the original Aligned label.
South Korea	0 Aligned	0.4 Significant	The 2021 NDC provides a specific 2030 transport-sector emissions level and an approximately 37.8% reduction from 2018, supported by clean-vehicle and demand-management measures. The IPCC reference is for 2050.	A strong and specific target is present, but a same-year pathway comparison is unavailable.
Türkiye	0 Aligned	0.4 Significant	The NDC includes dated passenger- and freight-rail modal-share targets and measures for rail investment, electrification and low-carbon transport. Modal share is not directly convertible to total transport emissions without a model.	The evidence supports a strong Significant screening label, not direct alignment.
United Kingdom	0 Aligned	0.4 Significant	The NDC and transport plan specify the phase-out of new petrol and diesel cars and vans by 2030 and zero tailpipe emissions for new cars and vans by 2035, with implementation measures. The target is vehicle sales, not total transport emissions.	The commitment is strong but does not justify a directly comparable Aligned label under the paper's D6 rule.

Entity	Original score/class	Audited score/class	Cut-off-compliant evidence summary	Source-audit rationale
United States	0 Aligned	0.6 Significant	The NDC describes vehicle-emissions standards, ZEV incentives and charging, transit investment, and low-carbon aviation and marine fuels, but does not state one quantified total transport-sector target.	The original Aligned label is too generous; Significant is the defensible screening outcome.

The full audit workbook, including official source URLs, evidence excerpts, confidence and rationales, is supplied as D6_Transport_17_Cell_Cutoff_Recheck.xlsx in Archive S1.

S15. Repository Contents and Reproducibility Status

Supplementary Archive S1 is an upload-ready review repository. In addition to the question, heatmap and D6 audit materials, the recovered archive contains 80 GPT-4o no-RAG outputs, 80 GPT-4o+RAG outputs, 80 Mistral+RAG outputs and 80 LLaMA+RAG outputs. It includes per-question claim counts for all four conditions and 579 one-row-per-claim no-RAG records with claim text, GPT-4o-mini label and judge reason. The RAG evaluation retained category counts but not the individual extracted claim strings or judge reasons; these unavailable RAG atomic records were not reconstructed.

Repository path	Purpose
data/questions/	80 questions, author-prepared references, reviewer answers and Full/Partial/None outcomes
data/heatmap/	Original 119-cell outputs, D6-audited reporting matrix, evidence passages, formal eight-cell and diagnostic 35-cell files
data/claims/	Per-question claim counts for all four model conditions and 579 atomic no-RAG claim records with labels and reasons
data/outputs/	Canonical 80-row outputs for GPT-4o no-RAG, GPT-4o+RAG, Mistral+RAG and LLaMA+RAG, plus original source JSON files
prompts/	Recovered RAG, claim-extraction, claim-verification and gap-analysis prompts; exact no-RAG prompt text was not found in the supplied code
scripts/	Classification metrics, heatmap checks, direct count-based paired bootstrap reproduction and repository validation
figures/	System architecture and original/revised heatmap figures
docs/	Codebook, known limitations, anonymisation and upload instructions

The recovered files support independent recomputation of the pooled point estimates and paired question-cluster bootstrap statistics. Individual RAG claim text and judge reasons remain unavailable because the original RAG evaluation serialized category counts only; these records were not reconstructed.